Diffusion-Forcing-Aware JEPA-Style VAE

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Abstract

Video diffusion models can generate high-fidelity rollouts(Ho et al., 2020; 2022; Rombach et al., 2022), but their performance in interactive environments depends critically on latent representations that preserve environment-specific details and action-conditioned temporal cues. Motivated by findings that transition-predictive objectives (modeling $p(s_{t+1} \mid \phi(s_t), a_t)$) yield strong representations for sequential decision making(Yang & Nachum, 2021; Nachum & Yang, 2021), we propose Diffusion-Forcing-Aware JEPA-VAE: a JEPA-style predictive objective applied to VAE latents(Assran et al., 2023), trained with teacher forcing to avoid compounding errors(Bengio et al., 2015). We integrate this representation learning into Diffusion Forcing Transformers (DFoT) for Minecraft video generation(Chen et al., 2024), jointly optimizing diffusion loss, JEPA latent prediction loss, and reconstruction regularization. We describe offline and online training paradigms and provide ablations comparing teacher forcing versus autoregressive latent rollouts, pairwise versus trajectory training, and frozen versus trainable VAE components.

1. Introduction

Learning from pixels in interactive 3D environments demands models that are simultaneously *generative* and *decision-relevant*. Minecraft is a particularly unforgiving testbed(Yan et al., 2023): small visual details and long-horizon action effects (navigation, mining, crafting) determine whether a rollout is plausible and whether a learned policy can act.

Modern pipelines often decouple these requirements by (i) compressing frames with a generic reconstruction-trained VAE(Kingma & Welling, 2014; Rombach et al., 2022) and (ii) training a sequence model in latent space. However, when representations discard action-conditioned cues, the downstream sequence model must compensate by implicitly re-learning state information that was never preserved.

A second, orthogonal challenge is *long-horizon stability* in continuous sequence generation. Auto-regressive next-token prediction is attractive because it supports variable-length rollouts and flexible conditioning, but teacher-forced training can become unstable on continuous signals; small errors accumulate and rollouts diverge when generating videos beyond the training horizon(Bengio et al., 2015). Full-sequence diffusion improves robustness for continuous data(Ho et al., 2020), yet typically relies on non-causal architectures and fixed-length generation. Diffusion Forcing (DF) bridges these regimes by training a causal denoiser with *independent noise levels per token*, combining flexible horizon generation with diffusion-style robustness and guidance(Chen et al., 2024; Song et al., 2025). This makes DFoT-style video transformers a compelling backbone for action-conditioned Minecraft rollouts.

In this paper, we argue that DF/DFoT's promise hinges on the *latent state definition*. If the VAE representation aliases distinct situations into the same latent code, no amount of sequence modeling can recover the missing information; indeed, aliasing is known to fundamentally break downstream imitation/control because different observations mapped to the same representation can require different actions(Nachum & Yang, 2021). Recent empirical and theoretical work suggests that representations optimized for transition prediction capturing $p(s_{t+1} \mid \phi(s_t), a_t)$ are particularly effective for sequential decision making(Yang & Nachum, 2021; Nachum & Yang, 2021). Complementing these empirical results, theory motivates learning representations that preserve action-conditioned transition structure by tying representation quality to errors in modeling rewards and transitions under offline data coverage.

Key idea. We make the VAE latent space explicitly *action-conditioned and future-predictive* by training it with a JEPA-style objective in latent space, while retaining decodability required for high-fidelity video diffusion. Concretely, we encode each frame into a latent state and train a causal predictor to predict the next latent state from the history of latent states and actions. To prevent collapse and trivial solutions, we use stop-gradient targets (predicting representations rather than pixels)(Grill et al., 2020; Chen & He, 2021; Assran et al., 2023) and anchor the representation with a reconstruction regularizer. Crucially, we train the predictor with *teacher forcing in latent space* to avoid compounding error during representation learning, while still enabling autoregressive latent rollouts as an ablation.

Diffusion-forcing awareness. DFoT trains with partial masking induced by independently noising tokens across time(Chen et al., 2024). A representation that is only predictive from fully clean history may be misaligned with DFoT sampling, where history can be partially degraded by noise. We therefore introduce an optional *noised-context JEPA* variant that perturbs the predictor's context latents with DF-style noise while predicting clean next-latent targets, aligning the representation objective with the sequence model's training and sampling conditions.

Contributions.

- We propose *Diffusion-Forcing-Aware JEPA-VAE*, a JEPA-style predictive objective applied to VAE latents to learn action-conditioned, future-predictive representations, addressing representation aliasing concerns for downstream control .
- We integrate JEPA-VAE training into DFoT-style latent video diffusion, leveraging Diffusion Forcing's per-token noising paradigm for robust long-horizon continuous rollouts.
- We introduce a diffusion-aware JEPA variant that trains predictiveness under partially noised history, matching DFoT's masking-by-noising regime.
- We present offline and joint-DFoT training protocols and a structured ablation suite (pairwise vs. trajectory training, teacher forcing vs. autoregressive latent rollouts, noised-context on/off, and freezing vs. finetuning VAE components).

2. Related Work

Latent video modeling and action-conditioned generation. A common strategy for scalable video generation is to compress frames with an autoencoder (e.g., a VAE)(Kingma & Welling, 2014; Rombach et al., 2022) and learn temporal dynamics in the latent space, which reduces computation while preserving perceptual fidelity. In interactive settings, models are typically conditioned on control signals (actions, goals, or plans), making the quality of the latent representation critical: if the encoder discards action-relevant details, the sequence model must implicitly recover them, often leading to brittle long-horizon rollouts.

Diffusion Forcing for sequences and video. Diffusion Forcing (DF) unifies next-token prediction and full-sequence diffusion by viewing both as *masking*, but along different axes: teacher-forced autoregression masks along time, while diffusion masks along noise.(Chen et al., 2024) DF trains causal sequence models to denoise sequences where each token can be assigned an *independent* noise level, enabling flexible horizon generation with diffusion-like robustness. This per-token noising provides a convenient mechanism for controlling how strongly the model "trusts" history versus treating it as corrupted context at sampling time, which is especially relevant for stabilizing long rollouts and for enabling guidance over future subsequences.

Representation learning for sequential decision making. A large body of work studies how offline representation learning can improve imitation learning and RL by training encoders with auxiliary objectives prior to (or alongside) policy learning.(Yang & Nachum, 2021; van den Oord et al., 2018; Srinivas et al., 2020; Nachum & Yang, 2021)

Empirically, objectives that predict components of a sub-trajectory from other components often via contrastive losses have been found to work particularly well across imitation, offline RL, and online RL. These results emphasize that learning representations from *state-action* context can matter more than pixel reconstruction alone, but also highlight practical pitfalls: naïve forward prediction in pixel space or poorly tuned latent forward models can underperform, motivating careful objective/architecture choices and ablations (e.g., whether to include actions, rewards, or momentum targets).

Connection to our approach. Our method sits at the intersection of these threads. We adopt a JEPA-style *predict-representations* objective in VAE latent space to explicitly encourage action-conditioned predictiveness, while retaining a reconstruction constraint to preserve decodability for latent video diffusion. We then make this representation learning *diffusion-forcing-aware* by optionally noising the predictor's context latents in the same spirit as DF's partial masking, aligning representation predictiveness with the conditions encountered by DFoT-style sampling.

JEPA and predictive representation learning. JEPA-style methods learn representations by predicting representations (not pixels), often using stop-gradient targets to prevent collapse and trivial solutions(Assran et al., 2023; Grill et al., 2020; Chen & He, 2021). We adopt this idea in latent space and couple it with reconstruction to preserve decodability.

3. Method

3.1. Overview

We aim to learn a latent representation that is simultaneously: (i) *decodable* for high-fidelity latent video generation with a diffusion-forcing transformer (DFoT), and (ii) *action-conditioned and dynamics-predictive* for sequential decision making. Our core idea is to treat the VAE latent as a *state*

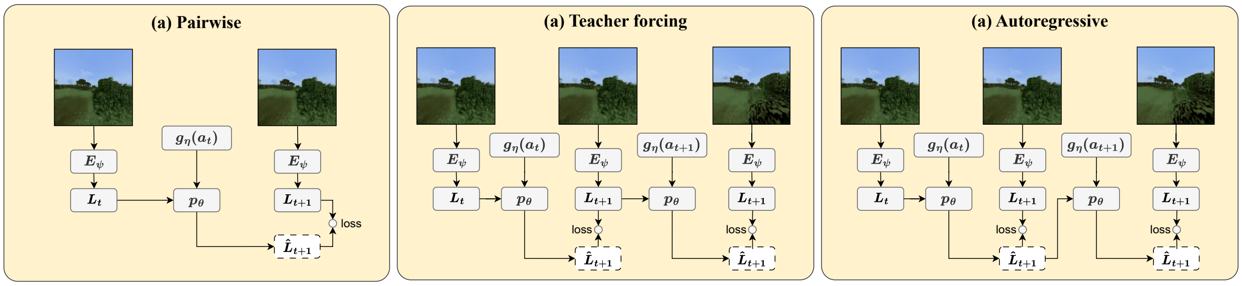

Figure 1. **Offline JEPA-VAE training strategies.** Each Minecraft observation o_t is encoded by a trainable VAE encoder E_ψ into a latent state L_t. Actions a_t are embedded by the action encoder $g_\eta(a_t)$ and consumed with L_t by a latent predictor p_θ to produce a one-step prediction $\hat{L}_{t+1}$ (dashed box). The JEPA objective matches predicted latents to encoder targets using a distance $d(\hat{L}_{t+1}, \mathrm{sg}(L_{t+1}))$, where $\mathrm{sg}(\cdot)$ denotes stop-gradient on targets. **(a) Pairwise:** optimize one-step transitions from sampled tuples (o_t, a_t, o_{t+1}). **(b) Trajectory teacher forcing:** unroll on trajectories while conditioning each step on the *target* latent L_{t+1} from the encoder (mitigates compounding error). **(c) Autoregressive rollout:** unroll by feeding back the *predicted* latent $\hat{L}_{t+1}$ to predict future steps (matches test-time rollout but may accumulate error).

and train it with a JEPA-style objective that predicts *future latents* from past latent–action context. The predictor is an auxiliary network used only to shape the encoder; video generation is still performed by DFoT in latent space.

3.2. Setup and notation

We consider trajectories $\tau = \{(x_t, a_t, m_t)\}_{t=0}^{T}$ where $x_t \in \mathbb{R}^{H\times W\times 3}$ is a frame, a_t is the action (discrete or continuous), and $m_t \in \{0, 1\}$ is a validity mask (padding/episode termination). We use a pretrained image VAE with encoder E_ψ and decoder D_ψ. The encoder defines a diagonal-Gaussian posterior

$$q_\psi(z_t \mid x_t) = \mathcal{N}\big(\mu_\psi(x_t), \mathrm{diag}(\sigma_\psi^2(x_t))\big), \tag{1}$$

where $z_t \in \mathbb{R}^{C\times h\times w}$ is a spatial latent tensor. For JEPA targets we use a deterministic latent (mode)

$$\bar{z}_t := \mu_\psi(x_t), \tag{2}$$

and define a flattened state vector $s_t := \mathrm{vec}(\bar{z}_t) \in \mathbb{R}^{d_s}$ with $d_s = C \cdot h \cdot w$. (For DFoT training we may use samples $z_t \sim q_\psi(\cdot \mid x_t)$, matching common latent diffusion practice; our implementation uses `sample()` for diffusion and `mode()` for JEPA targets.)

We embed actions via a learnable action encoder (MLP)

$$u_t := g_\eta(a_t) \in \mathbb{R}^{d_a}. \tag{3}$$

3.3. Latent dynamics predictor

We learn a causal predictor f_θ that maps a history of state-action pairs to next-state predictions:

$$\hat{s}_{t+1} = f_\theta(\{s_0 \parallel u_0, \ldots, s_t \parallel u_t\}), \quad t = 0, \ldots, T-1. \tag{4}$$

In practice, f_θ is a Transformer with causal self-attention (ViT-style blocks: LayerNorm $\to$ multi-head attention $\to$ MLP), operating over the *temporal* sequence of tokens. We first project s_t and u_t to a common hidden dimension, fuse them, add learned positional embeddings, and apply causal attention so position t can attend only to $\le t$.

Why a ViT/Transformer predictor? Minecraft latents are high-dimensional and multi-modal: the next latent depends on long-range temporal context (e.g., camera motion, inertia-like dynamics, inventory/UI changes) and on actions. Transformers are a natural choice because they: (i) model long-range dependencies with content-based attention, (ii) support variable-length conditioning windows, (iii) match the architectural bias of DFoT (also Transformer-based), making the auxiliary predictor compatible with the downstream temporal modeling stack. Importantly, f_θ is *not* used for final video generation; it is used to provide a strong, stable learning signal that shapes the encoder to preserve action-conditioned transition information.

3.4. JEPA-style latent prediction objective

We train E_ψ, g_η, and f_θ with a JEPA objective that predicts representations and matches them to stop-gradient targets:

$$\mathcal{L}_{\mathrm{JEPA}} = \mathbb{E}\left[\sum_{t=0}^{T-1} d\Big(\hat{s}_{t+1}, \mathrm{sg}(s_{t+1})\Big)\right], \tag{5}$$

Why teacher forcing? Our goal is to learn *better latents*, not to deploy f_θ as the rollout model. If we train f_θ autoregressively (feeding back $\hat{s}_{t+1}$), early prediction errors compound, and gradients to the encoder become dominated by "error amplification" artifacts rather than the true representational deficiency. Teacher forcing instead conditions each prediction on the *ground-truth* latent history $\{s_{\le t}\}$,

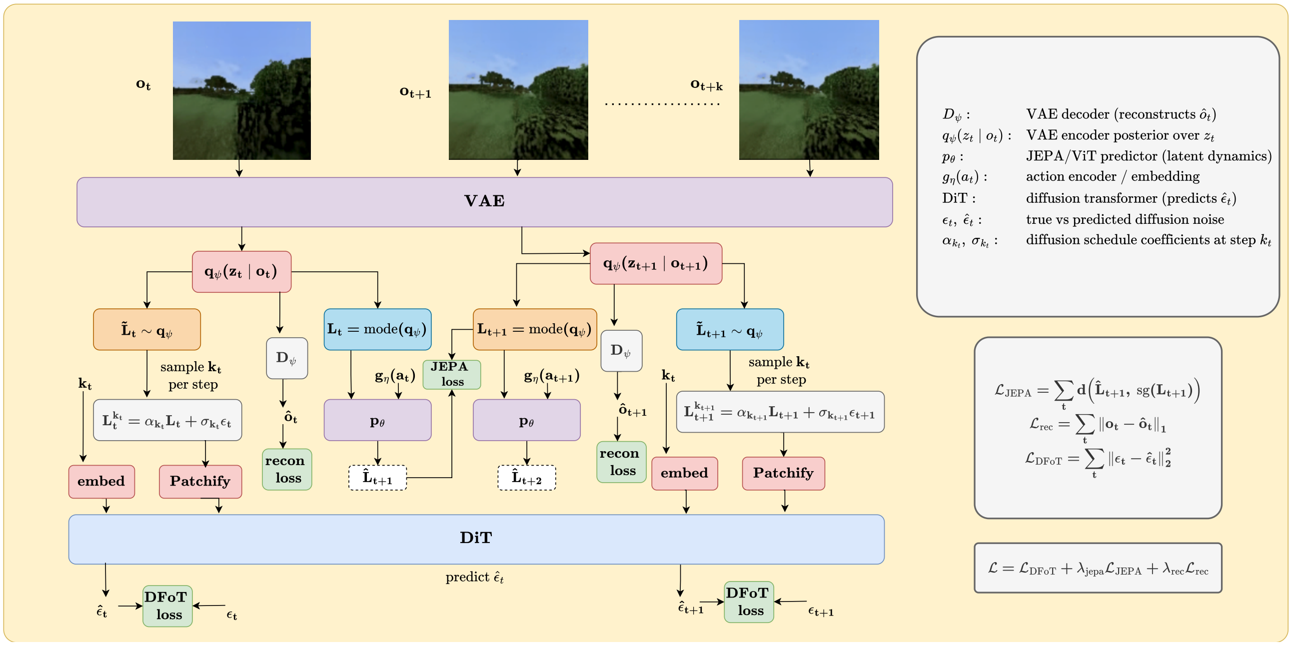

Figure 2. **Offline JEPA-VAE training strategies.** Each Minecraft observation o_t is encoded by a trainable VAE encoder E_ψ into a latent state L_t. Actions a_t are embedded by the action encoder $g_\eta(a_t)$ and consumed with L_t by a latent predictor p_θ to produce a one-step prediction $\hat{L}_{t+1}$ (dashed box). The JEPA objective matches predicted latents to encoder targets using a distance $d(\hat{L}_{t+1}, \mathrm{sg}(L_{t+1}))$, where $\mathrm{sg}(\cdot)$ denotes stop-gradient on targets. **(a) Pairwise:** optimize one-step transitions from sampled tuples (o_t, a_t, o_{t+1}). **(b) Trajectory teacher forcing:** unroll on trajectories while conditioning each step on the *target* latent L_{t+1} from the encoder (mitigates compounding error). **(c) Autoregressive rollout:** unroll by feeding back the *predicted* latent $\hat{L}_{t+1}$ to predict future steps (matches test-time rollout but may accumulate error).

yielding: (i) a well-conditioned supervised signal for "is s_t informative enough to predict s_{t+1} under action a_t?", (ii) stable gradients to E_ψ that encourage preserving transition-relevant details, (iii) a clean ablation axis: we still evaluate an autoregressive variant to quantify the gap, but we do not rely on it for representation shaping.

3.5. Reconstruction regularization (decodability constraint)

Predictive objectives alone can drift toward representations that are easy to predict but lose pixel-level information needed for video generation. We therefore regularize with reconstruction through the VAE decoder:

$$\mathcal{L}_{\mathrm{rec}} = \mathbb{E}\left[\sum_{t=0}^{T} \|D_\psi(\bar{z}_t) - x_t\|_1\right]. \tag{6}$$

In experiments we consider freezing vs. finetuning the decoder. Freezing D_ψ preserves the original VAE's reconstruction manifold, while finetuning can improve alignment with the JEPA-shaped encoder at the risk of drifting the diffusion latent space.

3.6. DFoT latent video diffusion objective

DFoT operates on latent sequences $z_{0:T}$ (latent tensors per frame). Let k_t be an independently sampled noise level for each time step, and define the per-token noising process

$$\tilde{z}_t = \alpha(k_t)\, z_t + \sigma(k_t)\, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I). \tag{7}$$

A causal diffusion-forcing transformer ϵ_ϕ predicts the noise (or an equivalent parameterization) given the noised sequence, noise levels, and optional conditions c (e.g., actions):

$$\mathcal{L}_{\mathrm{DFoT}} = \mathbb{E}_{z,\epsilon,k}\left[\sum_{t=0}^{T} \|\epsilon_t - \epsilon_\phi(\tilde{z}_{0:T}, k_{0:T}, c)\|_2^2\right]. \tag{8}$$

Independent $\{k_t\}$ makes training resemble "masking" tokens to different degrees; at sampling time, selectively noising history controls how strongly the model relies on past frames.

3.7. Diffusion-forcing-aware JEPA (future work)

A key mismatch can arise if JEPA trains predictiveness only from *clean* histories, while DFoT sampling may present *partially corrupted* histories due to per-token noising. To align the representation objective with DFoT conditions, we optionally noise the JEPA predictor's *context* states using the same (α, σ) schedule:

$$\tilde{s}_t = \alpha(k_t)\, s_t + \sigma(k_t)\, \xi_t, \qquad \xi_t \sim \mathcal{N}(0, I), \tag{9}$$

and predict clean next-state targets:

$$\hat{s}_{t+1} = f_\theta(\{\tilde{s}_0 \parallel u_0, \ldots, \tilde{s}_t \parallel u_t\}), \tag{10}$$

$$\mathcal{L}^{\mathrm{DF}}_{\mathrm{JEPA}} = \sum_{t=0}^{T-1} d\big(\hat{s}_{t+1}, \mathrm{sg}(s_{t+1})\big). \tag{11}$$

This trains latents to remain predictive even when history is partially "masked" by noise, mirroring DFoT's training/sampling regime.

3.8. Training regimes

Offline JEPA fine-tuning of the VAE (representation shaping before diffusion). We start from a pretrained VAE and fine-tune (some subset of) $\{\psi, \eta, \theta\}$ on an offline Minecraft trajectory dataset. The offline objective is

$$\min_{\psi,\eta,\theta} \; \lambda_J \, \mathcal{L}^{(\mathrm{DF})}_{\mathrm{JEPA}} + \lambda_R \, \mathcal{L}_{\mathrm{rec}}, \tag{12}$$

where $\mathcal{L}^{(\mathrm{DF})}_{\mathrm{JEPA}}$ is either the clean-context JEPA loss or the diffusion-forcing-aware (noised-context) variant. **Architecture:** only the VAE (encoder/decoder), action encoder, and predictor are used (no diffusion model). This stage produces a VAE latent space that is explicitly action-conditioned and dynamics-predictive, while remaining decodable.

Joint DFoT + diffusion-aware JEPA training (main setting). We jointly optimize the diffusion model and the JEPA-shaped representation so that the latent space is simultaneously good for denoising-based video generation and for action-conditioned predictiveness:

$$\min_{\phi,\psi,\eta,\theta} \; \mathcal{L}_{\mathrm{DFoT}} + \lambda_J \, \mathcal{L}^{(\mathrm{DF})}_{\mathrm{JEPA}} + \lambda_R \, \mathcal{L}_{\mathrm{rec}}. \tag{13}$$

4. Datasets and Experimental Details

4.1. Dataset: Minecraft video generation benchmark

We evaluate on the Minecraft dataset of Yan et al. (2023), consisting of action-labeled gameplay videos collected under a simple navigation policy. The agent executes one of three discrete actions at each time step: *forward*, *left*, and *right*. The dataset contains approximately 200K videos, each of length 300 frames, with an aligned action label per frame.

This benchmark is intentionally structured so that strong temporal metrics (e.g., FVD) require *long-range, action-conditioned* modeling. In particular, many trajectories involve turning and revisiting previously observed regions, making short-context generation insufficient even when per-frame fidelity is high. While the original dataset resolution is 128×128, we train and evaluate on an upsampled 256×256 version to target higher-quality generations.

Algorithm 1 Online DFoT + JEPA-VAE Training (Teacher Forcing)

1: Sample trajectory batch $\{(x_{0:T}, a_{0:T})\}$
2: Encode latents: $\bar{z}_t \leftarrow \mu_\psi(x_t)$; $s_t \leftarrow \mathrm{vec}(\bar{z}_t)$
3: Action embeddings: $u_t \leftarrow g_\eta(a_t)$
4: JEPA predict: $\hat{s}_{1:T} \leftarrow f_\theta(\{s_t \parallel u_t\}_{t=0}^{T-1})$
5: $\mathcal{L}_{\mathrm{JEPA}} \leftarrow \sum_{t=0}^{T-1}, d(\hat{s}_{t+1}, \mathrm{sg}(s_{t+1}))$
6: $\mathcal{L}_{\mathrm{rec}} \leftarrow \sum_{t=0}^{T}, \|D_\psi(\bar{z}_t) - x_t\|_1$
7: Sample per-frame noise levels $\{k_t\}$ and noise $\{\epsilon_t\}$; compute $\tilde{z}_t$
8: $\mathcal{L}_{\mathrm{DFoT}} \leftarrow \sum_{t=0}^{T}, \|\epsilon_t - \epsilon_\phi(\tilde{z}_{0:T}, k_{0:T}, c)\|_2^2$
9: Update parameters using $\mathcal{L}_{\mathrm{DFoT}} + \lambda_J \mathcal{L}_{\mathrm{JEPA}} + \lambda_R \mathcal{L}_{\mathrm{rec}}$

4.2. Training protocol

We train Diffusion-Forcing-Aware JEPA-VAE using the *joint diffusion-aware JEPA training* objective described in section 3, i.e., optimizing DFoT together with the JEPA loss (and optional reconstruction regularization) such that the learned latent space remains both decodable and action-conditioned/predictive.

Model configuration. Our DFoT backbone uses a DiT-style video transformer (`DiT/B`) with a 3D positional embedding scheme and patch size 2. The JEPA predictor is a causal Transformer (ViT-style blocks) with hidden size 512, depth 4, and 8 attention heads. Actions are embedded by a 2-layer MLP into a 256-dimensional embedding. For JEPA targets we use deterministic VAE latents (`mode()`) for stability; for diffusion training we follow standard latent diffusion practice and use stochastic latents (`sample()`) as implemented in our codebase.

Optimization details. We use AdamW with weight decay 10^{-3} and $\beta = (0.9, 0.99)$. The diffusion model learning rate is `1e`$-$`4`, the JEPA predictor/action-encoder learning rate is `1e`$-$`4`, and the VAE learning rate (when trainable) is `1e`$-$`5`. We use a constant-with-warmup schedule with 5K warmup steps. Unless otherwise stated, we set the JEPA loss weight $\lambda_J = 1.0$ and reconstruction loss weight $\lambda_R = 0.5$. We keep the VAE decoder frozen and finetune only the encoder (and quantization conv) to reduce latent-manifold drift.

Sequence lengths and context. For training and evaluation we generate sequences of $n\_frames = 8$ with a context length of 4 frames (history conditioning), i.e., the model predicts/generates 8-frame clips given the first 4 frames and action conditioning. Increasing these parameters was giving us out of memory error.

Compute budget and batch size. All experiments were run on $2\times$ NVIDIA A100 GPUs for approximately 16 hours. Due to memory constraints at 256×256 and the additional JEPA components, we used batch size 1 for training. This compute budget limited our ability to train from scratch or substantially extend training; therefore, we primarily study finetuning/joint training starting from pre-trained checkpoints.

4.3. Evaluation protocol and metrics

We report standard video generation and reconstruction metrics: FVD, FID, Inception Score (IS), LPIPS, MSE, PSNR, and SSIM. Following our compute constraints, we perform inference/evaluation on a limited subset of the dataset (50 videos total) and report metrics using this subset. This constraint primarily affects the variance of reported metrics; we use the same evaluation subset across methods for fair comparison.

5. Experiments and Results

We structure experiments around two hypotheses.

H1 (Representation matters). If VAE latents preserve action-conditioned transition information, DFoT should produce more temporally consistent rollouts (lower FVD) and more realistic videos (lower FID).

H2 (Alignment matters). Updating the latent space *without* co-adapting DFoT can hurt due to latent distribution/geometry shift. Therefore, offline JEPA-VAE fine-tuning alone may fail, while *joint diffusion-aware JEPA training* should succeed.

All reported metrics are computed under the same evaluation protocol from section 4 (8-frame clips with 4-frame context, action-conditioned). Lower is better for FID/FVD/LPIPS/MSE; higher is better for IS/PSNR/SSIM.

5.1. Offline JEPA-VAE fine-tuning (DFoT fixed)

Protocol. In the offline setting, we fine-tune the VAE encoder E_ψ (decoder frozen) and train the JEPA predictor f_θ and action encoder g_η *without* updating DFoT. At evaluation time, we *swap in* the offline-updated encoder and run generation with the same pretrained DFoT denoiser. This isolates whether “better latents” transfer to a fixed diffusion model.

Offline strategies. We compare: **Pairwise-TF**: one-step JEPA from tuples (x_t, a_t, x_{t+1}); **Rollout-TF**: trajectory JEPA with teacher forcing; **Rollout-AR**: trajectory JEPA with autoregressive predictor unrolling.

Why offline fine-tuning fails Offline JEPA changes the latent space. DFoT, however, is trained to denoise latents drawn from the *original* encoder distribution. When we replace the encoder but keep DFoT fixed, DFoT sees out-of-distribution latents and is forced to denoise in a geometry it never learned. This mismatch can dominate any representational improvement, producing worse FID/FVD even if the new latents are more predictive in principle.

Why Rollout-TF is best offline. Rollout-TF performs substantially better than Pairwise-TF/AR because it provides a multi-step training signal while keeping the conditioning history on the *ground-truth* latent trajectory. This tends to yield a smaller, smoother encoder update (less latent drift), so DFoT is less out-of-distribution at evaluation time. Even so, the remaining latent mismatch is enough to keep it worse than vanilla.

Why Pairwise-TF and Rollout-AR are especially harmful offline. Pairwise-TF provides only local constraints and can distort global latent geometry (scale/rotation/feature allocation) in ways DFoT is sensitive to, while still achieving good one-step predictiveness. Rollout-AR further injects compounding-error artifacts: the predictor’s early mistakes corrupt later contexts during training, and the encoder is shaped by gradients induced by these corrupted rollouts—an objective misaligned with what DFoT needs for denoising and sampling stability.

Takeaway. Offline JEPA-VAE fine-tuning alone does not transfer cleanly to diffusion-based generation. This motivates joint training where DFoT co-adapts to the evolving latent space.

5.2. Joint diffusion-aware JEPA training (DFoT + representation co-adaptation)

Methods and naming All methods start from the same pretrained DFoT+VAE initialization and fine-tune under the same compute budget. We compare:

- **Recon-FT (baseline):** fine-tune with $\mathcal{L}_{\text{DFoT}} + \lambda_R \mathcal{L}_{\text{rec}}$ (no JEPA).
- **JEPA-0R (ours):** fine-tune with $\mathcal{L}_{\text{DFoT}} + \lambda_J \mathcal{L}_{\text{JEPA}}$ and $\lambda_R = 0$.
- **JEPA+R (ours):** fine-tune with $\mathcal{L}_{\text{DFoT}} + \lambda_J \mathcal{L}_{\text{JEPA}} + \lambda_R \mathcal{L}_{\text{rec}}$.
- **JEPA-AR (ablation):** same as JEPA-0R but JEPA predictor trained autoregressively.

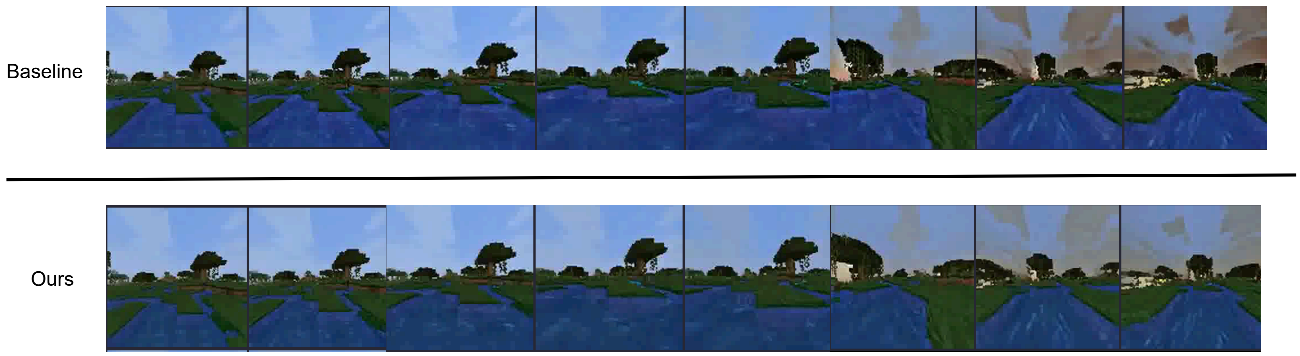

Figure 3. Baseline vs. our joint DFoT+JEPA model on the same conditioning. Joint JEPA training reduces long-horizon artifacts and improves temporal coherence across generated frames..

Method	FID ↓	FVD ↓	IS ↑	LPIPS ↓	MSE ↓	PSNR ↑	SSIM ↑
Recon-FT (baseline)	83.63	215.25	3.25	0.408	0.0175	17.56	0.499
JEPA-0R (ours)	**78.25**	**193.25**	**3.41**	**0.389**	0.0162	17.90	**0.516**
JEPA+R (ours)	80.63	212.63	3.26	0.402	**0.0161**	**17.94**	0.501
JEPA-AR (ablation)	88.56	264.25	3.21	0.420	0.0217	16.64	0.484

Table 1. **Joint diffusion-aware JEPA training.** JEPA-0R achieves the best FID/FVD; JEPA+R improves some pixel-space metrics but gives weaker temporal gains; JEPA-AR is unstable and degrades performance.

Key result: JEPA improves temporal quality (FVD) and realism (FID). JEPA-0R yields the best FVD (193.25) and best FID (78.25). This is the intended effect of our objective: JEPA encourages latents to retain the information needed to predict the future under actions, reducing representation aliasing across visually similar but dynamically distinct states. Because Minecraft contains many revisits/turnarounds where correct long-range evolution depends on action-conditioned state, this latent property directly improves long-horizon generation, which FVD is designed to capture.

Why JEPA-0R works well even without explicit reconstruction. At first glance, removing reconstruction seems risky because the encoder could drift toward non-decodable latents. In joint training, two forces still constrain decodability: (i) the decoder is frozen, so latents must remain in a region that can be decoded, and (ii) DFoT is trained to denoise latents that must decode to plausible frames, which penalizes arbitrary latent drift. This allows JEPA to reallocate representational capacity toward dynamics-relevant features without being tightly anchored to per-frame pixel reconstruction.

Why Recon-FT is a weak baseline for action-conditioned rollouts. Recon-FT optimizes per-frame fidelity but does not explicitly enforce action-conditioned predictiveness. As a result, the encoder can learn invariances that help reconstruct frames (or smooth nuisance variation) while still collapsing distinctions that matter for different futures under different actions. This mismatch between "what reconstructs well" and "what predicts well" is precisely what JEPA targets, explaining the gap between Recon-FT and JEPA-0R on FVD/FID.

Why JEPA+R improves pixel metrics but weakens temporal gains. JEPA+R achieves the best MSE/PSNR among JEPA variants, consistent with directly optimizing reconstruction. However, adding reconstruction reduces the FVD improvement relative to JEPA-0R. A principled interpretation is that reconstruction acts as a strong anchor on the latent manifold: it prevents representational reorganization that could improve transition sufficiency, especially when the decoder is frozen and compute is limited. Thus the hybrid objective exhibits the expected trade-off: better per-frame accuracy, weaker dynamics shaping.

Why JEPA-AR degrades performance. JEPA-AR performs worst among joint methods. Since the JEPA predictor is an *auxiliary* network used to shape the encoder (not the generator), autoregressive rollout introduces compounding-error artifacts into the training signal. These artifacts contaminate gradients into the encoder, encouraging stability under the predictor's own errors rather than producing a latent geometry that is easy for DFoT to denoise. Teacher forcing therefore provides a cleaner supervision signal for

Method	FID ↓	FVD ↓	IS ↑	LPIPS ↓	MSE ↓	PSNR ↑	SSIM ↑
Vanilla (no fine-tuning)	**82.44**	**196.75**	3.49	**0.372**	**0.0149**	**18.26**	**0.554**
Pairwise-TF	135.63	329.75	**3.45**	0.531	0.0245	16.10	0.500
Rollout-TF	88.75	263.75	3.37	0.405	0.0179	17.45	0.507
Rollout-AR	121.56	439.75	3.61	0.535	0.0248	16.05	0.490

Table 2. **Offline JEPA-VAE fine-tuning with a fixed DFoT denoiser.** Offline updating the encoder hurts generation quality relative to the vanilla pretrained DFoT+VAE pipeline. Rollout teacher forcing is the least harmful offline strategy but still worse than vanilla.

representation learning in this setting.

5.3. Key takeaways

- Offline JEPA-VAE fine-tuning does not work well for diffusion-based generation when DFoT is kept fixed, consistent with a latent distribution/geometry mismatch.

- Joint training resolves this mismatch and reveals the benefit of dynamics-predictive latents: JEPA-0R achieves the best temporal (FVD) and perceptual (FID) quality.

- Reconstruction regularization improves pixel-level fidelity but must be balanced carefully, as it can constrain representational reorganization needed for action-conditioned temporal consistency.

6. Conclusion

We presented **Diffusion-Forcing-Aware JEPA-VAE**, a representation learning approach that shapes VAE latents to be *action-conditioned and future-predictive* while remaining usable for latent video diffusion. Our key empirical finding is that *joint* diffusion-aware JEPA training improves action-conditioned Minecraft video generation, yielding substantially better temporal quality (FVD) and improved realism (FID) compared to a reconstruction-regularized latent adaptation baseline. In contrast, offline JEPA-VAE fine-tuning alone degrades generation, consistent with a latent distribution/geometry mismatch when swapping an updated encoder into a fixed pretrained DFoT denoiser.

Limitations. Our study is constrained by a modest compute budget (2×A100) and a limited evaluation subset, which restricts (i) training from scratch, (ii) extensive hyperparameter sweeps over loss weights and schedules, and (iii) scaling context lengths and sequence horizons. These constraints likely amplify trade-offs (e.g., between reconstruction anchoring and dynamics shaping) and may understate the potential of the full hybrid objective.

Future work. A natural next step is to strengthen the *diffusion-forcing awareness* of JEPA by matching the DFoT sampling regime more explicitly: e.g., applying per-token noise (or masking) to JEPA contexts using the same schedules as DFoT, conditioning JEPA on noised history frames, and exploring multi-step latent prediction objectives aligned with the denoising horizon. More broadly, integrating representation learning signals directly into diffusion training (beyond one-step prediction) may further improve long-range consistency and controllability.

Training from scratch: when might it help? Our ablations suggest that representation shaping is most valuable when the generative model can co-adapt to the latent space. Training DFoT *from scratch* jointly with JEPA-VAE is therefore especially promising in regimes where (i) the VAE latent space is not already well-aligned to the target domain, (ii) long-horizon action-conditioned consistency is critical (e.g., navigation with revisits, partial observability), and (iii) encoder drift would otherwise create a mismatch with a pretrained denoiser. In these settings, a from-scratch joint objective can learn a latent geometry that is simultaneously easy to denoise, easy to decode, and explicitly structured by action-conditioned dynamics precisely the inductive bias our method targets.

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